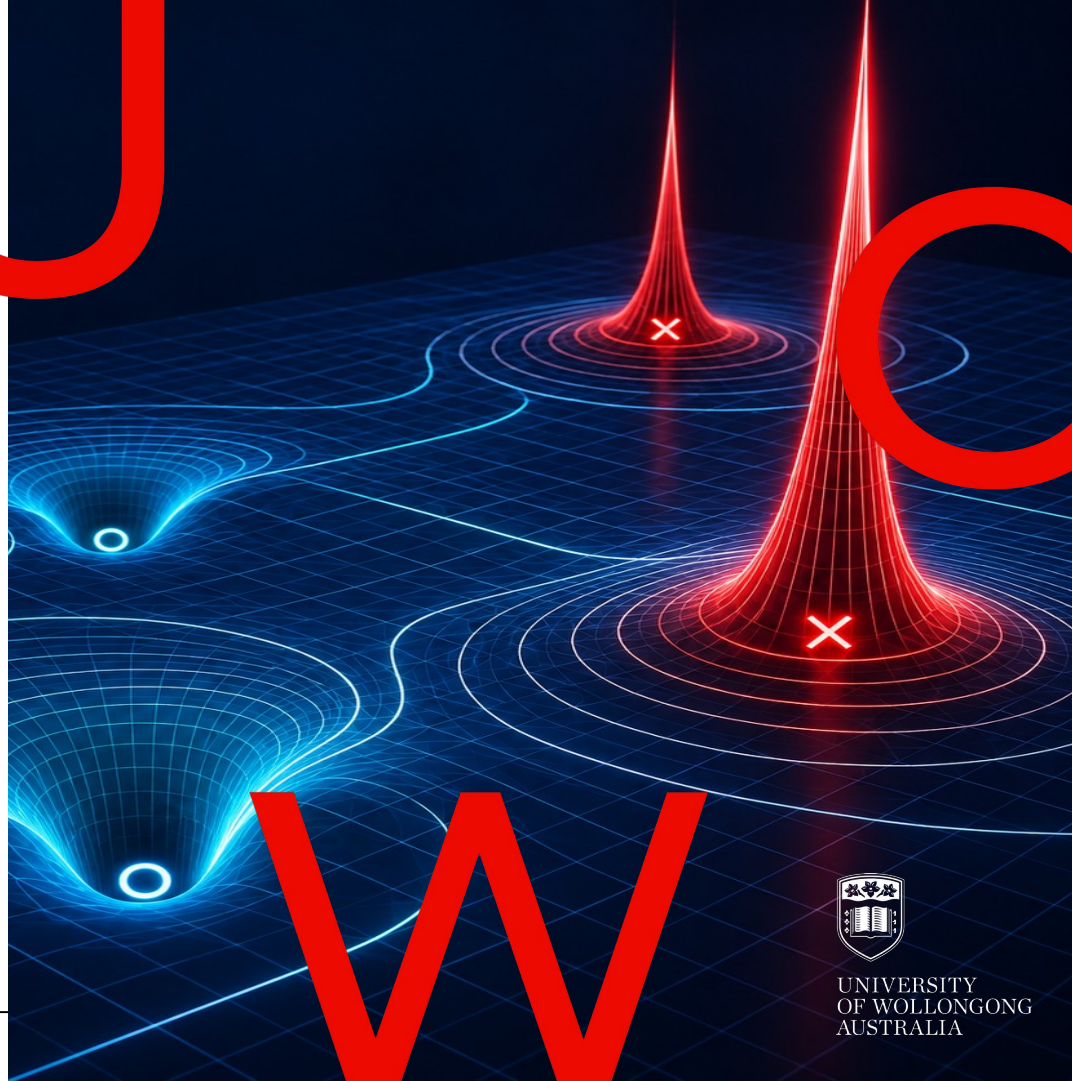
MECH321
Dynamics of Engineering Systems

Lecture 9 · Optional
Root Locus and Robustness

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Optional unit: gain selection, actuator limits and uncertainty.

Gain selection

OPTIONAL

Predict the response before choosing a gain.

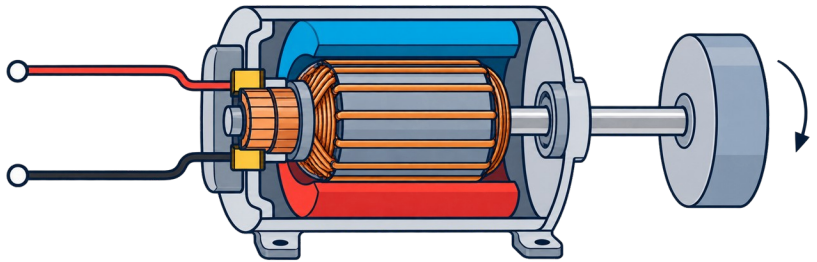
Position control of a robot joint

Tracking

For this position plant: smaller ramp error, smaller constant-load offset and a shorter first peak when $K > 1$.

Response and effort

Larger overshoot, higher initial actuator demand and less tolerance to omitted dynamics.



Electrical input → motor → shaft motion

These trade-offs are model-specific: a first-order speed loop need not have the same damping trade-off.

PART 1

Gain, poles and response

How does K move the closed-loop poles, and what response changes follow?

01 Derive the closed-loop roots

02 Read the root locus

03 Calculate and compare time responses

Position and speed models

OPTIONAL

Same actuator idea, different outputs and different system types

Normalised position model

$$\ddot{q}(t) + 2\dot{q}(t) = u(t) + d(t)$$

From rest. q is angle [rad]; u and signed load d are normalised acceleration commands [rad s^{-2}].

$$G_q(s) = \frac{Q(s)}{U(s)} = \frac{1}{s(s+2)}$$

$$u = K(r - q), \qquad [K] = \mathrm{s}^{-2}$$

Position adds an integrator. Stable P feedback is Type 1 with zero step-angle error. This lecture uses this model.

Speed model as a separate comparison

$$\omega = \dot{q}, \qquad G_\omega(s) = sG_q(s) = \frac{1}{s+2}$$

Type 0 with P. A 1 rad s^{-1} speed step with $K\omega$ = 4 s^{-1} gives:

$$e_{ss} = \frac{1}{1 + \frac{4}{2}} = \frac{1}{3}\ \mathrm{rad\,s^{-1}}$$

Project 2 uses physical v [V], ω [rad s^{-1}], τL [N m]; gain units V s rad^{-1}. Use the full Lecture 6 motor model and assessed data.

Position-loop gains and Type 1 step-error results are not numerical answers for a physical motor-speed model.

Characteristic equation

OPTIONAL

Gain moves roots even when the plant poles do not move

Closed loop

$$T(s) = \frac{KG(s)}{1 + KG(s)H(s)}$$

Poles

$$1 + KG(s)H(s) = 0$$

Meaning

Each K gives a different closed-loop polynomial and a different pole set

Workflow

calculate poles → predict response → check effort and robustness

Check: Open-loop poles belong to $G(s)H(s)$; closed-loop poles solve the characteristic equation.

Closed-loop roots: complete derivation

EXAMPLE

Solve the polynomial before comparing candidate gains

Derive the closed-loop roots

$$1 + \frac{K}{s(s+2)} = 0$$

$$s(s+2) + K = 0$$

$$s^2 + 2s + K = 0$$

$$(s+1)^2 = 1 - K$$

$$p_{1,2} = -1 \pm \sqrt{1-K}$$

Read stability and damping

$0 < K < 1$: $0 < \sqrt{(1-K)} < 1$, so both real roots are negative (overdamped).

$K = 1$: repeated root at −1 (critical).

$K > 1$: complex stable pair (underdamped).

$$p = -1 \pm j\sqrt{K-1}$$

$K = 0$: origin pole. $K < 0$: a positive real pole, hence unstable.

Closed-loop roots change with K; the open-loop plant poles remain at 0 and −2.

Pole movement with gain

EXAMPLE

Predict the damping type before reading the table.

Gain K	Closed-loop poles	Response
0.25	−0.134, −1.866	Overdamped
1	−1, −1	Critical
4	−1 ± j1.732	Underdamped
9	−1 ± j2.828	Underdamped

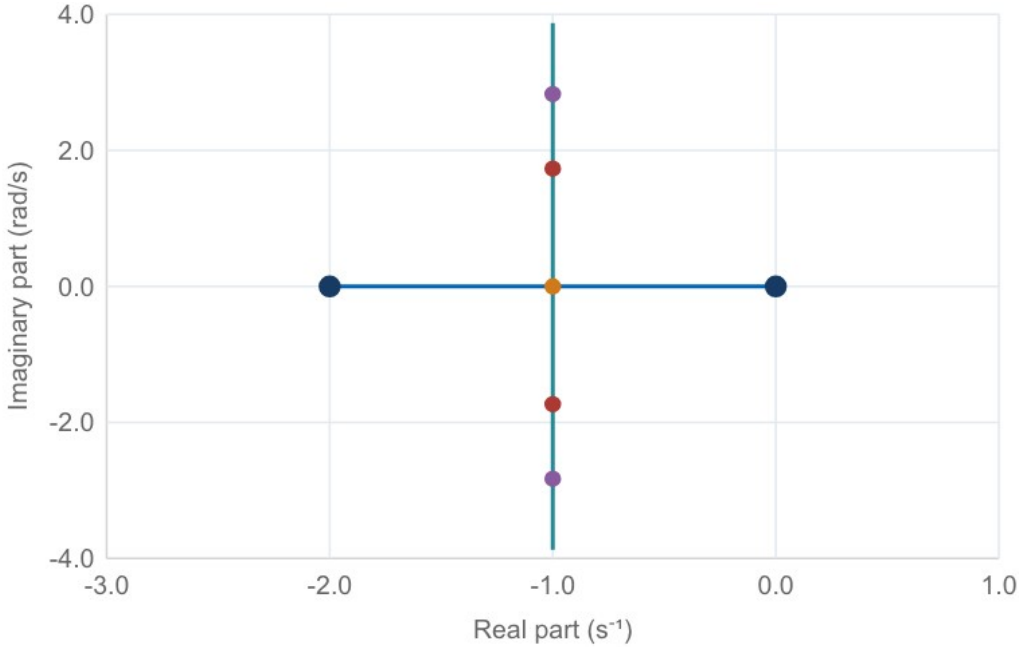
$$\zeta_{K=4} = 0.50, \qquad \zeta_{K=9} = 0.333$$

For $K > 1$, modal decay stays at 1 s^{-1}. The usual 2% settling estimate is about 4 s, not an exact invariant.

Root locus of the position loop

EXAMPLE

The gain scan becomes a continuous closed-loop pole trajectory



Reading the trajectory

At $K = 0$, the limiting poles are 0 and −2.
For 0<K<1, the real branches move towards −1.
At $K = 1$ they meet. For K>1 they move vertically at Re(p)=−1.
Orange: $K = 1$.
Red: $K = 4$, Im(p)=±1.732.
Purple: $K = 9$, Im(p)=±2.828.

CHECK Increasing K raises oscillation frequency here, while the modal decay rate stays unchanged.

From poles to two time responses

EXAMPLE

Calculate the speed, overshoot and decay separately for $K = 4$ *and* $K = 9$

$K = 4$: **moderate gain**

$\omega_n = \sqrt{4} = 2$ rad s^{-1}

$\zeta = \frac{1}{\sqrt{4}} = 0.50$

$\omega_d = \sqrt{4 - 1} = \sqrt{3}$

$t_p = \frac{\pi}{\sqrt{3}} = 1.81$ s

$M_p = e^{-\frac{\pi}{\sqrt{3}}} = 16.3\%$

$K = 9$: **high gain**

$\omega_n = \sqrt{9} = 3$ rad s^{-1}

$\zeta = \frac{1}{\sqrt{9}} = 0.333$

$\omega_d = \sqrt{9 - 1} = \sqrt{8}$

$t_p = \frac{\pi}{\sqrt{8}} = 1.11$ s

$M_p = e^{-\frac{\pi}{\sqrt{8}}} = 32.9\%$

Both have decay e^{-t} **and settling estimate ≈4 s. Faster first peak does not imply faster settling.**

Gain and response

EXAMPLE

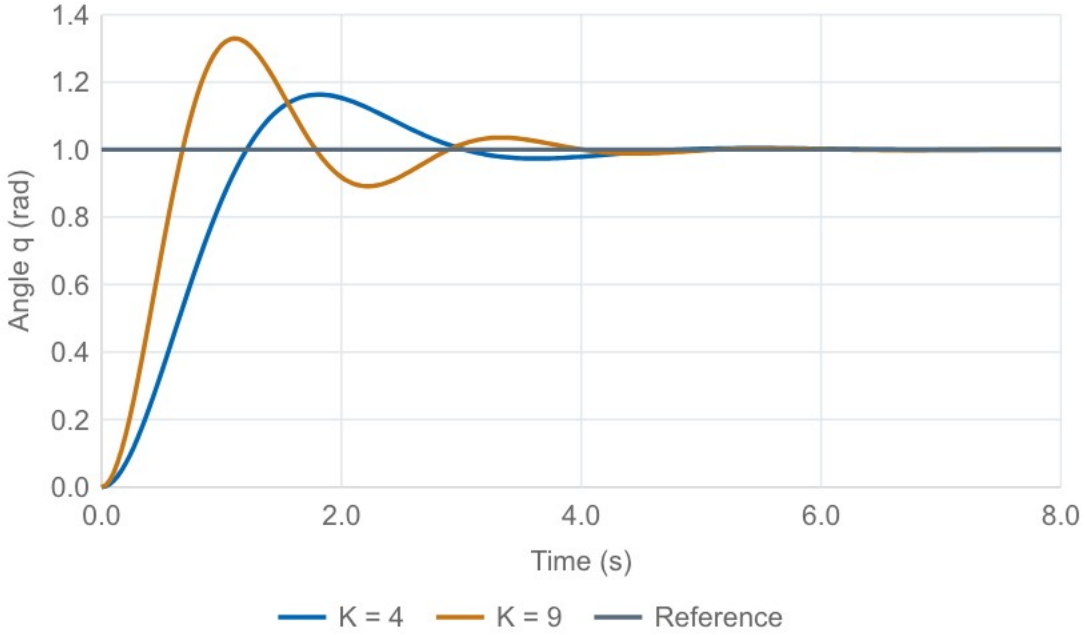
Prediction	$K=4$	$K=9$
Natural frequency	2 rad s^{-1}	3 rad s^{-1}
Damping ratio	0.50	0.333
First peak time	1.81 s	1.11 s
Overshoot	16.3%	32.9%
2% settling estimate	≈4.0 s	≈4.0 s
Initial effort (1 rad step)	4 rad s^{-2}	9 rad s^{-2}

$\zeta\omega_n = 1$ **for both. Read exact settling from the curve; higher gain also asks for higher actuator effort.**

Two step responses from the same model

EXAMPLE

Zero initial state, 1 rad *reference step, no load and no saturation*



Exact step solution

$$\omega_d = \sqrt{K-1}, \quad K > 1$$

$$q(t) = 1 - e^{-t}\cos(\omega_d t)$$

$$- \frac{e^{-t}\sin($$
$$\omega_d$$

$K = 4$: peak 1.81 s; 16.3%.
$K = 9$: peak 1.11 s; 32.9%.

Unit final angle. Both have the

The curves assume zero initial state, a 1 rad **step, no load and no saturation.**

PART 2

Nominal gain selection

Which gains satisfy tracking, step overshoot and actuator requirements together?

01 Tracking and effort

02 Intersect the nominal gain constraints

03 Check requested and delivered input

Control effort

OPTIONAL

A response improvement can require an impossible command

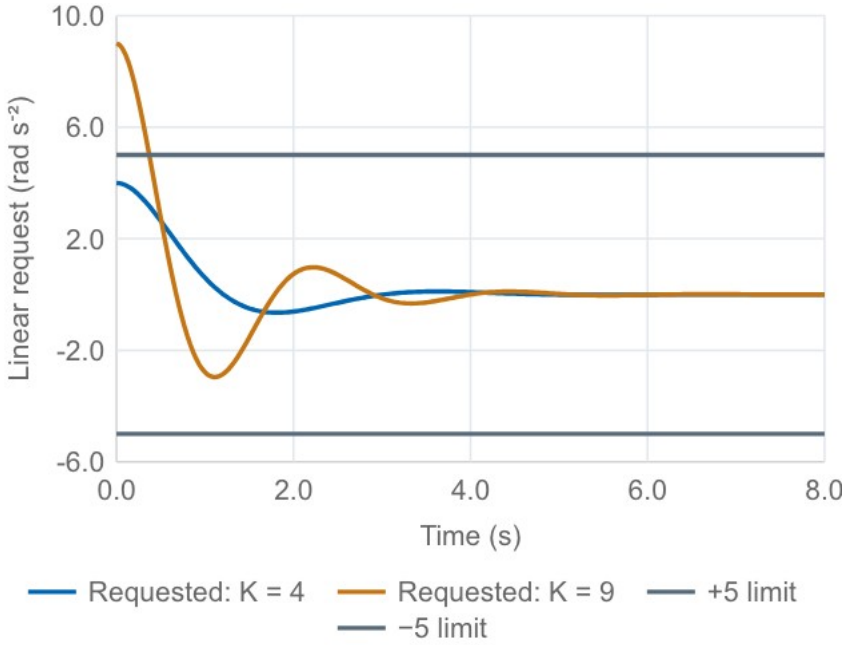
Check the actuator request

$$\frac{U(s)}{R(s)} = KS(s)$$

From rest, a 1 rad step gives $q(0^+) = 0$ and $e(0^+) = 1$ rad.

$$u(0^+) = K \text{ rad s}^{-2}$$

$K = 4$ requests 4 rad/s².
$K = 9$ requests 9 rad/s².
A ±5 rad/s² actuator cannot deliver the latter



These are unsaturated requests, not a simulation of delivered input. Always check u as well as q.

Tracking and actuator demand: full calculation

EXAMPLE

A ramp and a step test different requirements on the same gain

Ramp tracking

$$r(t) = vt, \qquad R(s) = \frac{v}{s^2}$$

$$E(s) = S(s)R(s) = \frac{(s+2)v}{s(s^2+2s+K)}$$

$$e_{ss} = \lim_{s\to 0} s\,E(s) = \frac{2v}{K}$$

For $v = 1$ rad s^{-1}: 0.50 rad at $K = 4$; 0.222 rad at $K = 9$. Final-value theorem requires a stable loop

Step startup

From rest, a 1 rad step gives $q(0^+) = 0$ and $e(0^+) = 1$ rad.

$$u(0^+) = Ke(0^+) = K \text{ rad}\,s^{-2}$$

With ±5 rad s^{-2}, $K = 4$ is delivered; the $K = 9$ request clips from 9 to 5.

Smaller ramp error does not justify $K = 9$.

State the command shape, amplitude and units before calculating an error or control limit.

Nominal admissible gain: derivation

EXAMPLE

Convert each requirement to an inequality before choosing a gain

Tracking and overshoot

$$\frac{2}{K} \le 0.60 \quad \Longrightarrow \quad K \ge 3.333$$

$$M_p = \exp\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right) \le 0.20$$

$$\zeta_{\min} = \frac{-\ln(0.20)}{\sqrt{\pi^2 + \ln^2(0.20)}} = 0.45595$$

Effort and intersection

A 1 rad step requests K rad s^{-2}.
Actuator limit: ±5 rad $s^{-2} \rightarrow K \le 5$.
Stability: $K > 0$.

$$\zeta = \frac{1}{\sqrt{K}} \quad \Longrightarrow \quad K \le \frac{1}{\zeta_{\min}^2} = 4.8102$$

$$3.333 \le K \le \min(4.8102, 5)$$

$$\boxed{3.333 \le K \le 4.8102}$$

Tests: 1 rad s^{-1} ramp and 1 rad step.

For $0 < K \le 1$, overshoot is zero but ramp error fails. The bounds use the stated commands and input limit.

Nominal gain decision

EXAMPLE

Requirements

1 rad s^{-1} ramp error ≤0.60 rad
1 rad step overshoot ≤20%
Step startup $|u|$≤5 rad s^{-2}
Closed-loop stable

Check each requirement before choosing K.

Tracking: $K \geq 3.333$
Overshoot: $K \leq 4.8102$
Effort: $K \leq 5$
Nominal interval: $3.333 \leq K \leq 4.8102$

$K = 4$ passes nominally: error 0.50 rad, overshoot 16.3%, startup $u = 4$ rad s^{-2}. Part 3 challenges this choice.

Saturation changes the dynamics

OPTIONAL

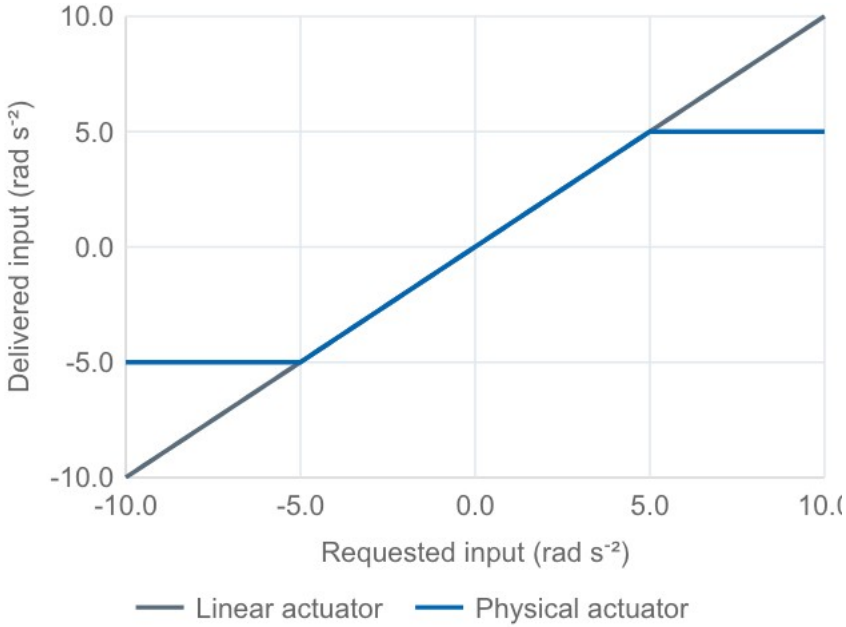
Linear poles apply only while the actuator remains linear

Requested ≠ delivered input

$$u_{\text{req}} = Ke$$

$$u_{\text{del}} = \text{sat}(Ke)$$

$$u_{\text{del}} = \begin{cases} -5, & u_{\text{req}} < \\ u_{\text{req}}, & |u_{\text{req}}| \\ 5, & u_{\text{req}} > \end{cases}$$



Feedback stays connected. The unsaturated poles do not describe clamped motion.

Report peak request, time clamped and recovery. Stable linear poles alone do not establish safe performance.

What saturation changes in this example

EXAMPLE

The actuator clamps the command, so the linear poles no longer govern startup

$K = 9$ **startup: the clamp is real**

$$e(0^+) = 1 \text{ rad}, \quad u_{\text{req}}(0^+) = 9 \text{ rad s}^{-2}$$

$$u_{\text{del}}(0^+) = 5 \text{ rad s}^{-2}$$

$$9e > 5 \quad \Leftrightarrow \quad e > \frac{5}{9} \text{ rad}$$

$$q < 1 - \frac{5}{9} = \frac{4}{9} \text{ rad}$$

Startup clamps while q stays below this threshold.

Simulate the physical feedback loop

Put the limiter before the plant, inside the feedback loop. Do not just clip a plotted linear response.

Record clamp duration, actual overshoot, settling time, and maximum rate or power demand.

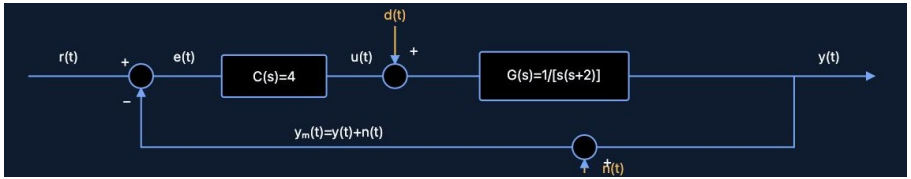
After leaving saturation, linear dynamics apply again. One linear pole set does not describe both phases.

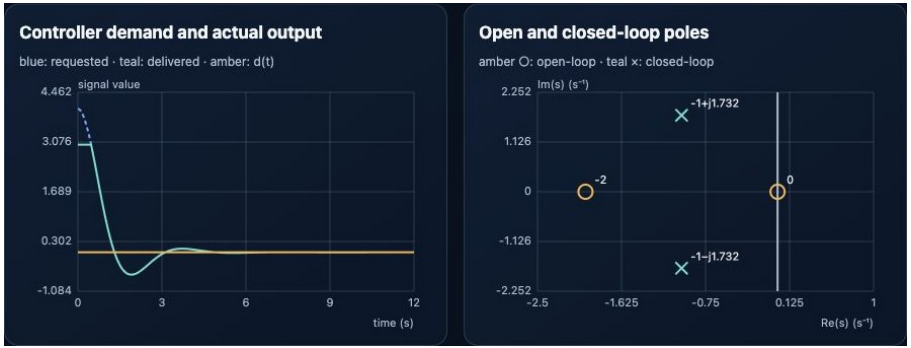
Feedback stays connected during saturation; reduced early acceleration can still give a poor recovery.

Gain and saturation

Predict the delivered input, then test the limiter inside the loop.

OPEN LIVE DEMO ↗





START Gain and saturation: Kc=4, $a = 2$, unit step, limit ±3 rad/s^2.

1 Toggle the limit. Compare u requested with u delivered.

2 Try Kc=9. Does more gain improve every response feature? Compare poles and $y(t)$.

MATLAB This code simulates the limiter in the governing ODE.

MATLAB alternative (base MATLAB)

```
K=4; U=3;
f=@(t,x)[x(2); ...
  min(U,max(-U,K*(1-x(1))))-2*x(2)];
[t,x]=ode45(f,[0 12],[0;0]);
u0=K*(1-x(:,1));
plot(t,[x(:,1),u0,min(U,max(-U,u0))]);
grid on; legend('y','requested','delivered')
```

Course home → Feedback Explorer → Gain and saturation

CHECK Initial request is 4 rad/s^2 but delivered control is 3. Linear poles remain the unsaturated prediction, not a model of clipped motion.

PART 3

Uncertainty and robustness

Does the same design satisfy every credible plant, not just the nominal model?

01 Separate S and T responsibilities

02 Test the adverse parameter limits

03 Challenge omitted dynamics

Sensitivity functions

OPTIONAL

S and T divide the loop's responsibilities

Definitions

$$S(s) = \frac{1}{1 + L(s)}, \quad T(s) = \frac{L(s)}{1 + L(s)}, \quad S(s) + T(s) = 1$$

Low frequency

Large $|L| \Rightarrow |S|$ small and $|T| \approx 1$

High frequency

Small $|L| \Rightarrow |T|$ small and $|S| \approx 1$

Meaning

One loop cannot make both S and T small at the same frequency

Check: Use S for output disturbance and model sensitivity; use T for reference tracking and sensor noise paths.

Plant-gain uncertainty

OPTIONAL

Test the full credible range

Fixed controller; uncertain plant

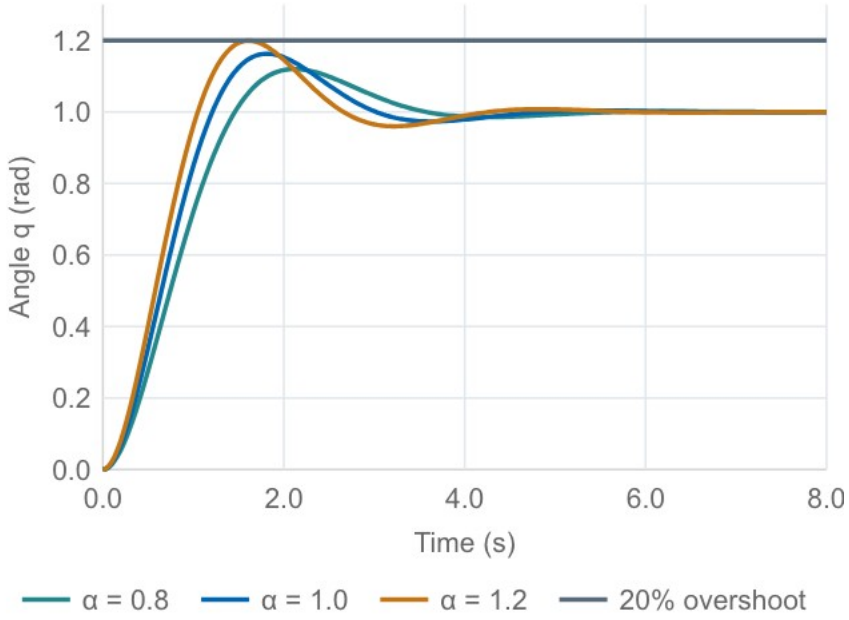
$$G_N(s) = \frac{1}{s(s+2)}$$

$$G_\alpha(s) = \alpha G_N(s), \quad 0.8 \le \alpha \le 1.2$$

$$K_{\text{eff}} = \alpha K, \qquad K = 4$$

$$3.2 \le K_{\text{eff}} \le 4.8$$

$$0.456 \le \zeta \le 0.559$$



High α is adverse for overshoot. Actuator request is still K e, not αK e.

Gain uncertainty

EXAMPLE

Which endpoint is worst for each requirement?

Low gain: worst ramp tracking

$$\alpha = 0.8, \quad K_{\text{eff}} = 0.8(4) = 3.2$$

$$e_{ss} = \frac{2}{3.2} = 0.625 \text{ rad}$$

This exceeds the 0.60 rad target for a 1 rad s^{-1} ramp. The nominal gain $K = 4$ fails the credible low-gain plant.

High gain: worst overshoot

$$\alpha = 1.2, \quad K_{\text{eff}} = 1.2(4) = 4.8$$

$$\zeta = \frac{1}{\sqrt{4.8}} \approx 0.456$$

$$M_p = e^{-\frac{\pi}{\sqrt{4.8-1}}} \approx 19.97\%$$

Close to the 20% step-overshoot limit. Recalculate both ends; do not call the nominal answer robust.

The adverse end is requirement-dependent: low α for ramp error, high α for overshoot.

Robust gain limits

EXAMPLE

The adverse tracking and overshoot bounds contradict one another

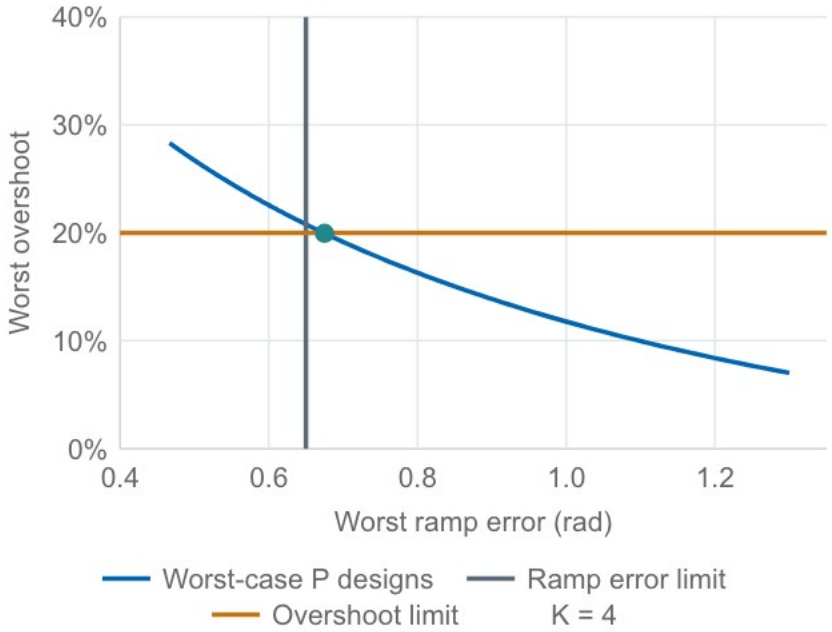
Tracking demands a larger K

$$\frac{2}{0.8K} \leq 0.60 \quad \Longrightarrow \quad K \geq 4.1667$$

Overshoot demands a smaller K

$$1.2K \leq 4.8102 \quad \Longrightarrow \quad K \leq 4.0085$$

Effort still requires $K \leq 5$.
No overlap: 4.1667 > 4.0085.
Change requirements or controller structure.



Acceptable designs must be left of 0.60 and below 20%. The P-gain curve never enters that region.

Neglected fast pole

OPTIONAL

A reduced model has a frequency limit

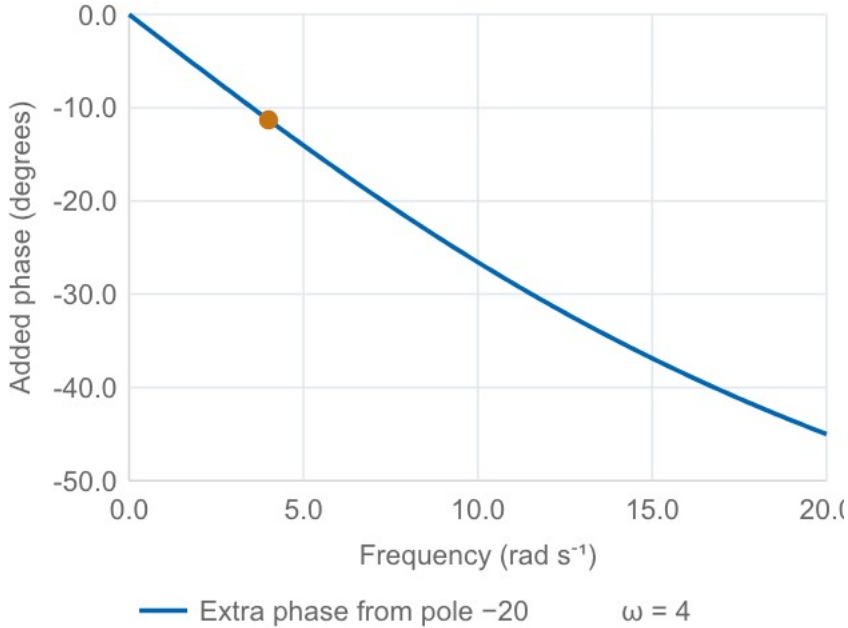
A fast pole is not invisible

$$G_r(s) = \frac{1}{s(s+2)}$$

$$G_f(s) = \frac{G_r(s)}{1 + \frac{s}{20}}$$

$$\Delta\phi(\omega) = -\tan^{-1}\left(\frac{\omega}{20}\right)$$

At $\omega = 4$ rad s^{-1}, the extra phase lag is 11.3°. A gain acceptable in the reduced model may lose margin in the full model



Lecture 8 uses the actual crossover frequency to quantify the phase-margin loss.

MATLAB verification of the gain decision

OPTIONAL / EXTENSION

Predict the poles and response before running the commands

Nominal pole and response check

```
s = tf('s');
G = 1/(s*(s+2));
K = 4; T = feedback(K*G,1);
pole(T); stepinfo(T);
rlocus(G);
t = (0:0.01:8)';
y = lsim(T,t,t);
e = t-y; u = K*e;
```

Uncertainty and interpretation

For a step, $u(t) = K[1 - q(t)]$. Plot this request against the actuator limit.
Repeat feedback(K*α*G,1) for α=0.8, 1 and 1.2.
Compare each ramp error and overshoot with the hand prediction.
Linear plots do not include input saturation. Put the limiter before the plant in a time simulation if the request exceeds it.

CHECK MATLAB verifies the stated model. It cannot repair an incorrect output choice or an omitted actuator limit.

Tutorial 9

Velocity feedback and uncertain damping

OPTIONAL

Optional Tutorial 9: pole design, uncertain damping and actuator limits

Q1: quick checks. Q2: design and MATLAB comparison. Q3: check a proposed revision against uncertain damping. Extension A: further practice.

Discuss before comparing answers

Explain why an acceptable initial command does not prove that the full actuator trajectory is feasible.

Optional study after Lecture 8; separate from Project 2.


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